

NeuroCR: Integrating Mixed Reality with Desktop Medical Imaging Software for Pre-surgical Evaluations of Epilepsy Patients

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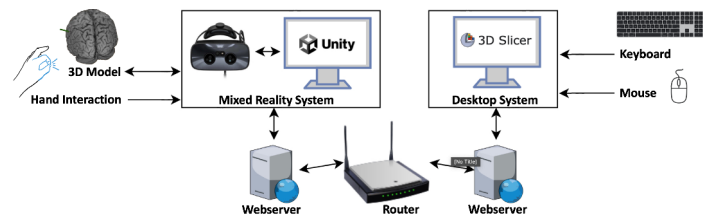


Figure 1: CR system architecture: The MR system (Unity application and HMD with hand tracking) and the desktop system (3D Slicer with mouse/keyboard inputs) communicate via networked webservers to synchronize 3D visualizations and interactions.

ABSTRACT

We present our prototype system, developed via a participatory design process with neurologists, demonstrating how Cross Reality technology can integrate the immersive, stereoscopic visualization and interaction benefits of eXtended Reality tools with traditional desktop medical imaging software. Our system enables clinicians to work simultaneously with both systems, with changes to the data synchronized between them. To address the clinical use-case, we adapted interaction techniques to make cross-sectional analysis more intuitive than on a flat screen. Feedback from the neurologists indicate a Cross Reality system has potential to enhance their current workflows.

Index Terms: Serious eXtended Reality, Mixed Reality, Cross Reality, Visualization, Visualization Techniques

1 INTRODUCTION AND RELATED WORK

Epilepsy often requires surgical intervention when other treatments fail. Accurate identification of epileptogenic zones—regions where seizures originate—is critical for surgical success [5]. This process relies on spatial Magnetic Resonance Imaging (MRI) data and temporal stereoelectroencephalography (sEEG) recordings, which neurologists traditionally analyze on 2D desktop monitors. However, these desktop-based tools have limitations in manipulating cross-sectional views and visualizing 3D brain structures, creating challenges in conceptualizing complex spatial data compared with the potentials of eXtended Reality (XR) tools [1]. XR technologies, including Mixed Reality (MR) and Virtual Reality (VR), offer improved depth perception and intuitive interaction with 3D data. Prior VR prototypes have reduced neurologists' cognitive burden and enhanced their understanding of higher-dimensional data compared to non-immersive desktop systems as well as assisted in multidisciplinary communication [1, 4]. However, VR systems often

lack the precision and familiarity of established desktop tools. CR systems address this challenge by seamlessly combining interfaces, allowing neurologists to interact with 3D models in MR while retaining access to desktop-based software for verification and documentation. Our system enables users to intuitively manipulate 3D MRI data using hand gestures, in a process similar to Botero-Ospina et al.'s touchless control module [3]; however, our system enables the user to work in an MR environment and synchronize changes with traditional desktop tools (see Figure 2). This hybrid approach offers promising enhancements to existing workflows, with the potential to improve both comprehension and efficiency during pre-surgical evaluations. We present a prototype Cross Reality (CR) medical image visualization system that integrates MR with traditional desktop-based tools to enhance neurologists' workflows during pre-surgical evaluations of epilepsy patients.

2 PARTICIPATORY DESIGN PROCESS

We collaborated with three neurologists experienced in pre-surgical evaluations of epilepsy patients to inform the system design. Elicitation sessions were held to describe how CR technologies can combine XR with desktop tools, and garner expert opinion of how CR could enhance their workflows. The neurologists emphasized the importance of controllerless, hand-based interactions to improve intuitiveness and streamline transitions between MR and desktop tools. They emphasized the importance of prioritizing visualization and manipulation of brain cross-sections, and also proposed a novel interaction method to anchor cross-sections around an sEEG electrode axis (see Figure 3: A-B).

3 SYSTEM ARCHITECTURE AND TECHNOLOGIES

The system comprises two interconnected subsystems: the MR application and the desktop software, which communicate via WebSockets. The desktop system uses a modified version of 3D Slicer (a medical imaging platform) to send and receive updates when users reposition cross-sections. The MR system, developed in Unity 2022 LTS and run on the Varjo XR-3 head-mounted display (HMD), synchronizes changes with the desktop software, ensuring consistency between views. The MR system supports volumetric rendering of NIfTI-formatted MRI files using the Unity asset VolumeViewerPro (now deprecated) and incorporates Ultraleap hand tracking for intuitive, controllerless interactions. Users can manipulate 3D models and cross-sections using predefined hand gestures:

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grab (move the 3D visualization), pinch (adjust cross-section positions), and point (interact with menu controls and electrodes). They can also slice a cross-section at angles defined by the rotation of their palm, with a technique similar to the Handsaw [2], however unlike the Handsaw, our method allows for slicing at any angle.

3.1 Mixed Reality User Interface

To access settings and toggle features of the system, an augmented menu is anchored to the users palm, which is made visible when they rotate their palm towards their eyes. The menu has buttons enabling users to show and reposition the cross-sectional planes (axial, coronal, sagittal), anchor an oblique cross-section to an electrode, and dynamically adjust cross-section orientation using hand gestures. Additional controls modify the visualization's transparency, contrast, and brightness to suit task-specific needs.

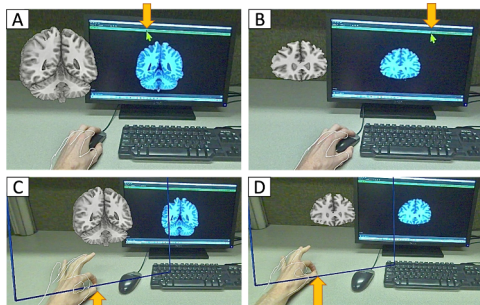


Figure 2: Bi-directional CR interactions: (A-B) A mouse in 3D Slicer updates the cross-section view in both desktop and MR systems (arrow highlights movement of mouse cursor). (C-D) A "pinch" gesture in MR updates the view, synchronized across both systems (arrow highlights movement of the hand gesture).

4 FEEDBACK AND FUTURE WORK

To evaluate the initial prototype, one-hour demonstration sessions were conducted with neurologists. Each session included an introduction to the system, followed by hands-on testing during which participants were encouraged to think aloud. Follow-up questions were posed after testing. Participants appreciated the intuitive hand-gesture controls for repositioning 3D models and cross-sections, as well as the ability to freely grab and rotate visualizations, which enhanced their spatial comprehension of anatomical relationships. The MR resolution was considered sufficient for clinical use, though suggestions were made to enable zooming into specific regions without resizing the entire model. One neurologist found the CR system superior to VR, citing an increased sense of grounding from being able to see both the desktop monitor and their physical surroundings. They also noted that combining MR with traditional systems enhanced their understanding of the "network" of data. Another neurologist appreciated the convenience of being able to move freely and utilize their workspace while interacting with the visualization. However, limitations in hand-gesture recognition were noted, attributed to variations in hand shapes and dexterity. Furthermore, the resolution of the desktop screen via MR video pass-through was insufficient for analyzing fine details, such as sEEG waveforms, leading to a recommendation for integrating high-resolution virtual desktop overlays into the MR display. In followup discussion after testing, the neurologists had future considerations that included supporting visualization of the multiple contact points on individual electrodes, animating seizure progression linked to sEEG recordings, and enabling overlay of imaging modalities like positron emission tomography (PET) or single photon emission computed tomography (SPECT) to identify overlapping problem areas. The neurologists also emphasized the value of a multi-user system to facilitate real-time collaboration, such as

when they are presenting their evaluations to other clinical staff. In addition to these suggestions, we determined future work should address scalability, automated detection of electrode positions, and generalizing the system for broader clinical applications.

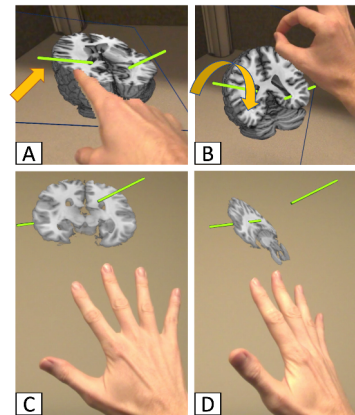


Figure 3: Interactions: (A-B) User "points" to anchor cross-section around selected electrode, then "pinches" to rotate the cross-section around the electrodes axis (arrow highlights the selected electrode). (C-D) Users control the cross-section orientation via the position and rotation of their palm, enabling intuitive and dynamic slicing through the 3D model (arrow highlights axis of rotation around electrode).

5 CONCLUSION

We developed a CR prototype that integrates MR with traditional desktop medical imaging tools to enhance neurologists' workflows during pre-surgical evaluations of epilepsy patients. By synchronizing MR immersive visualizations with desktop views, the system enables intuitive manipulation of imaging data using both hand gestures and traditional inputs simultaneously. Novel features, such as anchoring cross-sections around sEEG electrodes or having cross-section angles defined by the rotation of the user's palm, simplify complex slicing tasks. The ability to easily grab and rotate the entire visualization in MR delivered an improved understanding of spatial relationships. Limitations include the need for improved hand gesture precision and reliability as well as improving the resolution of the desktop screen through the MR headset. Further iterations of the prototype will allow for user-studies to further assess system design and usability.

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